

0.1 p -Groups

Theorem 0.1.1. Let G be a group of order p^n for some prime p and $n \in \mathbb{N}$. Then, G has a normal series $\{P_i\}_{i=0}^n$ s.t

$$1 = P_0 \triangleleft P_1 \triangleleft \cdots \triangleleft P_{n-1} \triangleleft P_n = G$$

where $|P_i| = p^i$ for each i .

Proof. First, we recall that for any p -group G , the center $Z(G) \neq 1$. This follows from the class equation:

$$p^n = |G| = |Z(G)| + \sum |G : C_G(g_i)| = |Z(G)| + \sum p^{r_i} \quad (1 \leq r_i \leq n-1).$$

This implies p divides $|Z(G)|$, so $|Z(G)| > 1$.

Now, we use induction on n . For $n = 2$, $|G/Z(G)| = p$ or 1 . In either case, $G/Z(G)$ is cyclic, which implies G is Abelian. By Cauchy's Theorem, G has a subgroup of order p , and the claim is satisfied.

For $n \geq 2$, suppose the claim is true for groups of order p^n . Let $|G| = p^{n+1}$. Since $1 \subsetneq Z(G)$, by Cauchy's Theorem, there exists a normal subgroup $N \triangleleft G$ such that $N \leq Z(G)$ and $|N| = p$. Now, G/N is a group of order p^n . By the induction hypothesis, G/N has a normal series:

$$1 = N_0/N \triangleleft N_1/N \triangleleft \cdots \triangleleft N_{n-1}/N \triangleleft N_n/N = G/N.$$

The i th Isomorphism Theorem gives $N_{n-1} \triangleleft G$ and $|N_{n-1}| = p^n$. Thus, by induction, the claim is proved. $\square$

Lemma 0.1.2. $G/Z(G)$ is cyclic $\iff G$ is Abelian.